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A Retrieval-Augmented Generation (RAG) Based Framework for Evaluating Urban Low-Carbon Governance and Its Implications for Sustainable Development

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Abstract

The transition to low-carbon urban environments is a cornerstone of global sustainability efforts. However, evaluating the diverse management efforts driving this transition is frequently hindered by fragmented and heterogeneous data. This study introduces a novel Retrieval-Augmented Generation (RAG) framework to systematically evaluate urban low-carbon governance. RAG represents a significant methodological innovation, integrating advanced information retrieval with generative artificial intelligence to provide a transparent and evidence-based assessment of policy implementation. The proposed framework is applied to evaluate the low-carbon energy practices of 296 Chinese cities. A critical finding is the systemic neglect of the "Check" phase across all urban tiers, posing a significant challenge to the long-term sustainability of carbon reduction initiatives. Conversely, high-performing cities are characterized by robust "Feedback" mechanisms. Policy priorities also vary significantly with city scale: larger cities emphasize strategic development and low-carbon transitions, while smaller cities focus on foundational planning and ecological preservation. This framework serves as a transparent, process-oriented tool for evidence-based low-carbon governance, thereby facilitating more resilient and sustainable urban pathways.

Keywords: low-carbon development; policy evaluation; retrieval-augmented generation; energy policy; Chinese cities; PICOF cycle

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1. Introduction

Global climate change poses an unprecedented challenge to the long-term sustainability of human society, necessitating rapid and deep decarbonization. China, as a major emitter, has established ambitious national strategies for low-carbon development, aiming to peak carbon emissions by 2030 and achieve carbon neutrality by 2060 [1]. City governments are pivotal in this transition, acting as key implementers of national directives and drivers of local innovation. While extensive research has assessed the outcomes of urban low-carbon initiatives, such as emissions reductions and renewable energy adoption [2,3], a critical gap remains: the systematic evaluation of the governmental efforts that underpin these outcomes. An outcome-oriented approach often fails to recognize substantial efforts by cities that face significant contextual challenges, thereby overlooking valuable policy in-

novations and hindering equitable cross-regional learning, which is essential for achieving balanced regional sustainability.

Evaluating these governmental efforts necessitates a deep analysis of vast and varied textual data, including policy documents, official reports, and public communications. This documentation provides a rich narrative of policy intentions, implementation strategies, and monitoring mechanisms. However, traditional methods of policy analysis are ill-equipped to handle the scale and complexity of this data. Manual analysis is labor-intensive, prone to inconsistencies, and difficult to scale across numerous cities and administrative bodies [4]. While computational text analysis has been applied in policy studies, it often fails to integrate with established governance evaluation frameworks, leaving a gap in process-oriented assessment.

Although sharing similarities with traditional models like Plan-Do-Check-Act (PDCA) [5], the PICOF framework offers distinct advantages for this research. Specifically, it differentiates 'Implement' (the execution of policy) from 'Outcome' (the measurable results), and explicitly includes a 'Feedback' loop, which is critical for understanding adaptive governance in long-term policy cycles like low-carbon transitions. This nuanced structure provides a more granular and realistic lens for evaluating governmental efforts compared to the more generalist PDCA cycle. However, systematically applying PICOF to large-scale textual data has been methodologically challenging. This disconnect creates a significant methodological gap, limiting our ability to conduct transparent, replicable, and process-oriented evaluations of public policy, which are vital for ensuring the accountability and sustainability of governance practices.

To bridge this methodological gap, this paper introduces a novel framework that utilizes a Retrieval-Augmented Generation (RAG) system to systematically evaluate the low-carbon development efforts of city governments. RAG, an advanced AI paradigm, combines the deep contextual understanding of large language models with the factual grounding of targeted information retrieval. This synergy provides a powerful tool for analyzing extensive document corpora with high accuracy and consistency. Our RAG-based approach is specifically designed to operationalize the PICOF policy cycle, enabling a structured and nuanced assessment of governmental efforts in the critical energy sector. This approach directly supports sustainable urban governance by providing a scalable and objective means to monitor and optimize policy performance.

This study makes several significant contributions. First, it introduces a methodological innovation by applying RAG in environmental policy evaluation, presenting a new frontier for analyzing complex governmental efforts with unprecedented scale and precision. Second, it develops a holistic evaluation framework by integrating the PICOF cycle, facilitating a comprehensive, process-oriented assessment that transcends outcome-focused metrics to understand the entire policy implementation lifecycle. Third, it provides empirical insights through a comparative analysis of 296 Chinese cities, identifying patterns and disparities in low-carbon energy policy efforts and offering actionable knowledge for policymakers to enhance urban sustainability.

The remainder of this paper is structured as follows: Section 2 reviews the literature on low-carbon city evaluation and policy analysis techniques, highlighting the limitations that motivate our RAG-based approach. Section 3 details our methodological framework, including the RAG architecture and evaluation criteria. Section 4 presents the empirical findings from our analysis of Chinese cities. Section 5 discusses the implications of our findings, and concludes with limitations and future research directions.

2. Literature Review

2.1. Low-Carbon City Evaluation

Evaluating low-carbon initiatives at the city level has gained significant attention in recent years, as urban centers are critical to achieving global sustainability targets [6]. Researchers have developed various methodological approaches to assess urban climate action, which can be broadly categorized into indicator-based systems, qualitative assessments, and hybrid methodologies.

Indicator-based evaluation frameworks represent the most prevalent approach in the literature. For instance, Tan et al. [7] developed a comprehensive index system comprising indicators spanning economic, social, and environmental dimensions to evaluate low-carbon city performance in China. Similarly, Han et al. [8] proposed an ecological security pattern framework composed of multi-level ecological elements, based on ecosystem services, human impacts, and improved connectivity, emphasizing the ecological dimension of urban sustainability.

Qualitative assessment methods, while less common, provide valuable insights into the contextual factors and governance processes underlying low-carbon transitions. Hewa et al. [9] identify common constraints hindering the delivery of low-carbon buildings in high-rise high-density cities and utilize social network analysis based on expert input to map the specific stakeholder engagement required to overcome these challenges. These qualitative evaluations often reveal nuanced implementation challenges that may not be captured by quantitative indicators alone, highlighting the socio-political complexities of sustainable development.

Hybrid methodologies attempt to combine the strengths of both approaches. For example, Li et al. [10] employed the TOE framework and fuzzy-set qualitative comparative analysis (fsQCA) on 35 key Chinese cities to explore multiple paths for urban low-carbon governance. Similarly, Guo et al. [11] utilized qualitative comparative analysis (QCA) and a case study of a Chinese pilot city to determine effective low-carbon policy implementation strategies.

Despite these methodological advances, existing evaluation approaches face critical limitations. First, quantitative data scarcity remains a significant barrier, particularly for smaller cities or within specific sectors. Second, the prevalent focus on outcomes over processes fails to account for varying baseline conditions and implementation challenges, creating an incomplete picture of governmental performance. Most importantly, these methods lack the capability to systematically analyze and scale the vast corpus of textual policy documents that detail the efforts themselves. This limitation highlights the need for a new approach that can bridge the gap between qualitative policy richness and quantitative analytical rigor, thereby providing a more robust foundation for sustainable urban management.

2.2. Policy Evaluation Techniques

Policy evaluation in the environmental and energy domains has evolved significantly, drawing on diverse methodological traditions from public administration, economics, and environmental sciences. These evaluation techniques can be categorized based on their temporal focus (ex-ante vs. ex-post), methodological approach (quantitative vs. qualitative), and evaluation criteria such as effectiveness, efficiency, and equity.

Ex-ante evaluation techniques aim to assess potential policy impacts before implementation. Cost-benefit analysis (CBA) represents a widely used approach in energy policy evaluation, as demonstrated by Boateng et al. [12], who performed a cost-benefit analysis on China's Smart Grid investment from 2020 to 2050, revealing substantial economic benefits crucial for integrating renewable energy and achieving net-zero goals. Similarly,

multi-criteria analysis (MCA) has been employed to evaluate energy poverty, as shown in Siksnelyte et al.'s [13] assessment across EU countries in the context of low-carbon transitions, which is a vital aspect of social sustainability.

Ex-post evaluation methods assess policy outcomes after implementation. Econometric approaches, such as difference-in-differences and regression discontinuity designs, have been used to estimate the causal effects of energy efficiency policies, as exemplified by Jin et al. [14] in their analysis of building heating energy in Chinese cities. Furthermore, Lin et al. [15] used a difference-in-differences model on panel data to demonstrate that China's electricity price subsidy policy effectively and sustainably mitigated NO_X emissions from coal-fired power plants by incentivizing the installation of denitrification units.

In the specific context of environmental and energy policy, several specialized evaluation frameworks have emerged. Environmental impact assessment (EIA) methodologies have been adapted for policy evaluation, as demonstrated by Nita et al. [16]. Sousa et al. [17] conceptualize a multi-level policy framework and a corresponding evaluation framework to guide innovation policies toward fostering sustainable socio-technical transformations by addressing potential failures and assessing both policy outputs and social impacts, thereby aligning policy interventions with broader sustainability objectives.

Despite this diverse toolkit, policy evaluation in the energy domain remains constrained. Attribution problems, time lags, and the complexity of multi-level governance pose significant challenges. Furthermore, conventional frameworks often struggle to systematically incorporate the rich, qualitative data embedded in policy texts, which is essential for understanding the *process* of implementation. This gap underscores the need for innovative methods that can effectively analyze large volumes of unstructured textual data to provide a more holistic and timely assessment of policy efforts, ensuring that governance remains responsive and sustainable.

2.3. RAG Technology

Retrieval-Augmented Generation (RAG) represents an emerging paradigm in artificial intelligence that combines the strengths of information retrieval systems with generative language models. First introduced by Lewis et al. [18], RAG architectures enhance the capabilities of large language models by grounding their outputs in relevant retrieved information, thereby improving factual accuracy and domain specificity.

The core architecture of RAG systems consists of two primary components: a retriever module and a generator module. The retriever identifies relevant documents or passages from a knowledge base in response to a query, while the generator produces coherent text that incorporates both the query and the retrieved information. This architecture offers several advantages over pure generative models, including improved factual grounding, greater transparency through citation of sources, and enhanced domain adaptation without requiring full model retraining, making it a robust tool for evidence-based sustainability research.

Initial applications of RAG technology focused primarily on question-answering tasks. For example, Soto et al. [19] used semantic retrieval and language models to generate reliable answers from domain-specific knowledge bases (tested on the pollution domain). Similarly, Siriwardhana et al. [20] applied RAG to career advising question-answering, showing improved accuracy in domain-specific responses.

More recently, RAG applications have expanded beyond simple question-answering to more complex analytical tasks. Ramadhan et al. [21] employed RAG for document summarization, extracting malnutrition information from electronic health records in Iraqi aged care facilities. Xiao et al. [22] utilized prompt engineering for segmenting long

documents and generating structured summaries, demonstrating RAG's potential for long document analysis.

In the specific context of policy analysis, RAG technology offers several promising capabilities for enhancing urban sustainability. A practical study by Chang et al. [23] evaluated open-source LLMs in local RAG systems for low-carbon data applications in China, demonstrating the feasibility of using such systems for domain-specific analysis. RAG technology can process large volumes of heterogeneous policy documents more efficiently than traditional methods, identify relevant policy segments across different document types and governance levels, and generate structured evaluations based on specific analytical frameworks, maintaining consistency across multiple cases.

Despite its potential, the application of RAG in public policy evaluation, particularly within environmental and energy governance, is nascent. This research gap presents a compelling opportunity to pioneer the use of RAG as a robust methodological tool to systematically analyze policy documents, thereby enhancing our understanding of low-carbon policy implementation and overcoming the limitations of traditional evaluation techniques, ultimately contributing to more effective and sustainable policy design.

3. Methodology

3.1. Data Acquisition and Preprocessing

The empirical analysis of this study is based on a comprehensive corpus of policy-related documents from 296 prefecture-level and above cities in China for the year 2023. To ensure data representativeness and balance, we constructed a multi-source dataset, which is essential for a reliable assessment of urban sustainability. The primary data source consists of official policy documents, including government work reports, five-year plans, and specific low-carbon or energy-related policy directives sourced from the official websites of municipal governments. All documents are in Chinese and primarily in PDF or HTML formats. In total, 1360 such documents were collected based on a keyword search strategy focusing on terms like "energy structure," "low-carbon," and "green development." We applied strict quality control, filtering out duplicate or irrelevant texts to ensure the corpus reflects authoritative government stances. To supplement this, we also collected 164,431 news articles and official press releases from authoritative local media outlets to capture the public communication and real-time progress of policy implementation. For the 'Outcome' dimension, quantitative data, such as energy consumption statistics, were sourced from the China Urban Statistical Yearbook and provincial statistical yearbooks. All collected textual documents underwent a standardized preprocessing pipeline, including text extraction, format normalization, and cleaning, to prepare them for the knowledge base construction as detailed in Section 3.3. This rigorous data preparation ensures that the subsequent RAG-based analysis is grounded in high-quality, relevant information, thereby enhancing the validity of our sustainability evaluation.

3.2. Evaluation Variables and Metric Construction

To systematically evaluate the efforts of city governments, we adopt the "Plan-Implement-Check-Outcome-Feedback" (PICOF) framework, which conceptualizes policy implementation as a continuous management cycle [24]. As illustrated in Figure 1, this framework allows for a structured assessment of the entire policy process: from initial planning (Plan) and execution (Implement) to monitoring (Check), results assessment (Outcome), and adaptive adjustment (Feedback). This cyclical perspective provides a holistic lens through which to analyze governmental efforts in the energy sector, offering a dynamic view of how cities evolve towards sustainable energy systems.

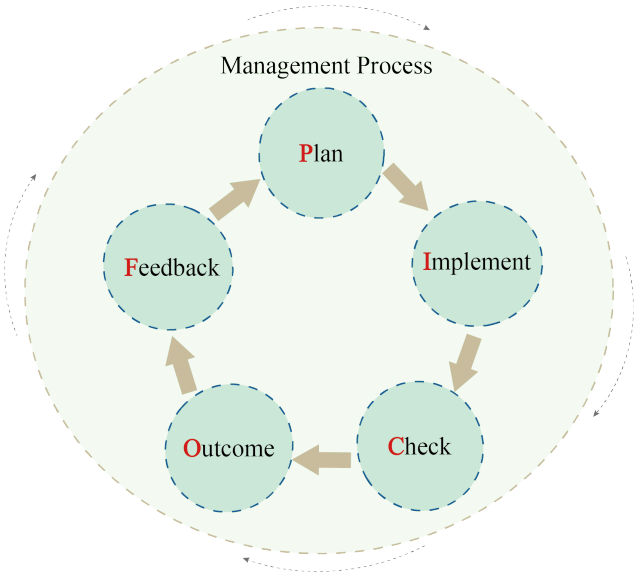


Figure 1. Illustration of PICOF circle in management process. Note that the first letters of the five phases (P-I-C-O-F) are highlighted to represent the PICOF acronym. For the quantitative evaluation, each phase concludes with a calculated score, serving as a termination point for the assessment period rather than an infinite loop.

Based on this management process, we defined the specific evaluation variables shown in Table 1 to measure the efforts score of city governments in the energy dimension, aligning them with the multifaceted requirements of sustainable urban development.

Table 1. PICOF evaluation variables for urban low-carbon governance.

Phase	Evaluation Variables
Plan (P)	1. Non-fossil energy development and application planning objectives. 2. Non-fossil energy development and application planning content.
Implement (I)	1. Degree of improvement in relevant implementation regulations. 2. Investment in special funds for optimizing energy structure. 3. Human resource support for optimizing energy structure. 4. Technical condition support for optimizing energy structure.
Check (C)	1. Degree of improvement in relevant supervision regulations. 2. Special fund guarantee for supervision. 3. Human resource support for supervision. 4. Technical condition support for supervision.
Outcome (O)	1. Proportion of non-fossil energy in primary energy consumption (%). 2. Proportion of coal in primary energy consumption (%).
Feedback (F)	1. Incentives for entities that reduce emissions by optimizing energy structure. 2. Penalties for entities that fail to reduce emissions due to not optimizing energy structure. 3. Further measures and plans to achieve emission reduction through optimization of energy structure.

For each evaluation variable within the P, I, C, and F phases, we developed detailed scoring criteria and formulated them into specific prompts. These prompts guide the RAG system in its semantic analysis and quantitative scoring of policy documents. For the Outcome phase, quantitative data was sourced from official publications like the China Urban Statistical Yearbook and subsequently normalized for scoring.

Our RAG-based framework quantifies the textual information from policy documents into discrete scores through precise semantic matching. These scores, combined with the normalized scores from the quantitative Outcome data, are aggregated using predefined weights: Plan (0.2), Implement (0.2), Check (0.15), Outcome (0.3), and Feedback (0.15). The

allocation of these weights was determined through a Delphi method involving ten experts from the fields of energy policy and urban governance, two of whom possess interdisciplinary backgrounds in artificial intelligence and machine learning. This consensus-based weighting ensures that the evaluation reflects the strategic importance of each phase for long-term sustainability while accounting for the technical characteristics of the evaluation model. The experts rated the relative importance of each PICOF phase, and the final weights represent the averaged consensus, reflecting the greater emphasis on measurable outcomes and foundational actions while still valuing monitoring and adaptation. This balanced weighting is crucial for evaluating the comprehensive nature of sustainable governance.

The scoring mechanism is operationalized as follows. First, a comprehensive knowledge base is constructed from municipal policy documents concerning energy and low-carbon development. For each evaluation variable, a specific prompt is engineered to query this knowledge base. The RAG system then retrieves the most relevant document segments and generates an analysis-based score according to predefined criteria. Scores for the Plan, Implement, Check, and Feedback phases reflect policy comprehensiveness, specificity, and innovation. For the Outcome phase, quantitative indicators are normalized to ensure comparability across diverse urban contexts.

This integrated mixed-methods framework enables a balanced and comprehensive assessment, evaluating both the procedural efforts (qualitative policy analysis via RAG) and the performance results (quantitative energy indicators) of a city's low-carbon initiatives. By bridging the gap between inputs and outputs and incorporating expert validation, this approach provides a more nuanced evaluation of urban sustainability than traditional methods. It thus facilitates a deeper understanding of the mechanisms driving sustainable energy transitions in urban settings.

3.3. Evaluation Framework

The core of our methodology is the RAG-based evaluation framework, depicted in Figure 2. This framework is designed to systematically process and analyze textual data from official documents, assessing the low-carbon efforts of various cities. The workflow is divided into two main stages: knowledge base construction and RAG-driven information extraction and scoring.

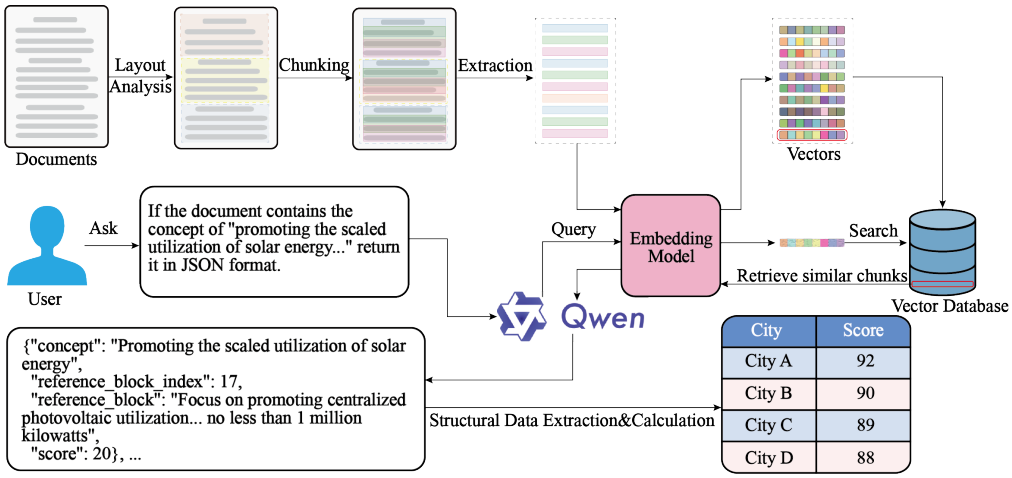


Figure 2. Illustration of RAG-based evaluation framework.

1. Knowledge Base Construction: A comprehensive corpus of relevant documents, including government policies, official reports, and news articles, pertaining to low-carbon development in Chinese cities is collected. These documents then undergo a rigorous preprocessing pipeline. First, layout analysis and text extraction are performed; documents,

often in PDF or Word format, are parsed to identify structural elements and extract raw text content while discarding irrelevant components. Second, the extracted text is segmented into smaller, semantically coherent chunks to enhance the precision of subsequent information retrieval. Third, each text chunk is transformed into a high-dimensional vector representation using the BGE M3-Embedding model [25], which was selected for its state-of-the-art performance on Chinese language semantic retrieval tasks and its ability to capture the semantic essence required for sustainability assessments. Finally, the generated embeddings, along with their corresponding text chunks and essential metadata, are indexed and stored in a specialized vector database optimized for efficient similarity searches.

2. RAG-based Information Extraction and Scoring: The information extraction and scoring process follows a structured workflow. For each evaluation variable under the PICOF framework, specific natural language queries are engineered to elicit information directly relevant to the variable being assessed. During the retrieval phase, the query is embedded into the same vector space as the document chunks. A hybrid retrieval strategy combines dense vector-based semantic similarity with sparse keyword-based matching (utilizing the BM25 algorithm for sparse retrieval and cosine similarity for dense retrieval, combined through a weighted scoring mechanism), ensuring that the retrieved chunks are both semantically related and contain precise key terms, thereby facilitating a more accurate and comprehensive evidence-based evaluation of sustainability efforts. The system retrieves the top-K (where K = 15) most relevant text chunks to serve as the context.

In the generation phase, the query and retrieved contextual chunks are fed into the Qwen-max large language model. This model was chosen for its strong performance in Chinese language understanding, ensuring consistency across the 296 cities. The LLM is prompted to act as a policy analyst and generate a structured response, typically in JSON format, which encapsulates key extracted information and preliminary scores. Finally, the information extracted from all relevant documents for a given city is aggregated to compute a final score for each variable, synthesized into an overall low-carbon construction capability score. This evidence-driven approach enhances the objectivity and depth of urban sustainability analysis by grounding it in verifiable policy documentation.

This RAG-based framework enables a nuanced, evidence-driven evaluation by directly grounding the assessment in the content of official documents, thereby enhancing the objectivity and depth of the low-carbon practice analysis.

3.4. Framework Validation

To evaluate the reliability and consistency of the RAG-based evaluation framework, we conducted a validation experiment comparing its results with manual assessments. A random sample of 20 cities was selected from our dataset. Two independent researchers, both with expertise in Chinese energy policy, were tasked with manually scoring the PICF dimensions for these 20 cities based on the same corpus of policy documents and the scoring criteria outlined in Section 3.2. The inter-rater reliability between the two manual coders was high, with a Cohen's Kappa of 0.85, indicating substantial agreement. We then calculated the Pearson correlation coefficient between the RAG-generated scores and the averaged manual scores for each of the PICF dimensions. The results showed strong positive correlations across all dimensions: Plan (r = 0.88), Implement (r = 0.86), Check (r = 0.91), and Feedback (r = 0.89). These high correlation coefficients suggest that our RAG-based framework produces results that are highly consistent with expert human judgment, thus validating its reliability and robustness as a policy evaluation tool. In addition to this quantitative validation, we performed a qualitative spot-check. Leveraging the visual frontend of the RAGflow framework used in this study, three researchers randomly sampled and reviewed approximately 300 individual scoring records generated by the

system. The review found that 91% of the automated scores were deemed reasonable and well-supported by the retrieved textual evidence, further confirming the framework's practical accuracy.

4. Results

4.1. Analysis of Aggregate Low-Carbon Performance

This study evaluated the low-carbon energy development levels of 296 Prefecture-Level-and-Above (PLAA) cities in China—excluding Sansha due to data unavailability—using the RAG-based approach to score the PICF dimensions, which is a critical step towards monitoring urban sustainability. Concurrently, statistical data for the O (Outcome) dimension were collected and transformed into corresponding scores. A composite performance score for each city was derived by calculating a weighted sum of the five dimensions (PICOF), employing weights of 0.2, 0.2, 0.15, 0.3, and 0.15, respectively. The statistical distribution of these final scores, illustrated in Figure 3, approximates a normal distribution, with scores ranging from a minimum of 16.50 to a maximum of 71.62.

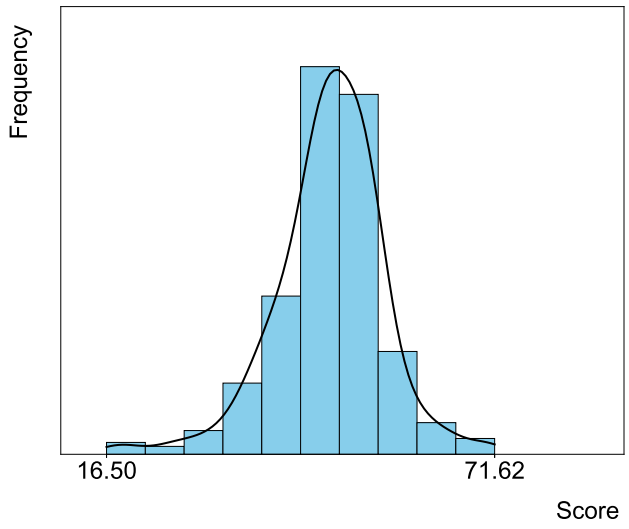


Figure 3. Distribution of the overall performance scores.

Based on these results, cities were classified into four performance tiers using equal intervals to represent distinct levels of low-carbon construction: Tier A (57.84–71.62), Tier B (44.06–57.84), Tier C (30.28–44.06), and Tier D (16.50–30.28). We selected the equal interval method over relative grouping methods (such as quantiles or clustering) to establish an absolute benchmark for performance evaluation. This approach ensures that a city's tier depends solely on its own score rather than the performance of others, thereby facilitating longitudinal monitoring and setting clear, fixed targets for policy improvement. This tiered classification provides a structured framework for assessing the relative sustainability of urban energy practices across China.

Geographical disparities in these scores were visualized through a geospatial distribution map, as presented in Figure 4. The analysis reveals that the 19 Tier A cities are predominantly concentrated in China's eastern and southern coastal regions, with significantly lower representation in the western and northern provinces, reflecting regional imbalances in sustainable development.

In accordance with the classification criteria established by the Chinese State Council (2014), the 296 PLAA cities are categorized into 21 mega cities (Group I), 80 large cities (Group II), 111 medium cities (Group III), and 85 small cities (Group IV). Table 2 summarizes the distribution of cities across performance tiers for each scale group.

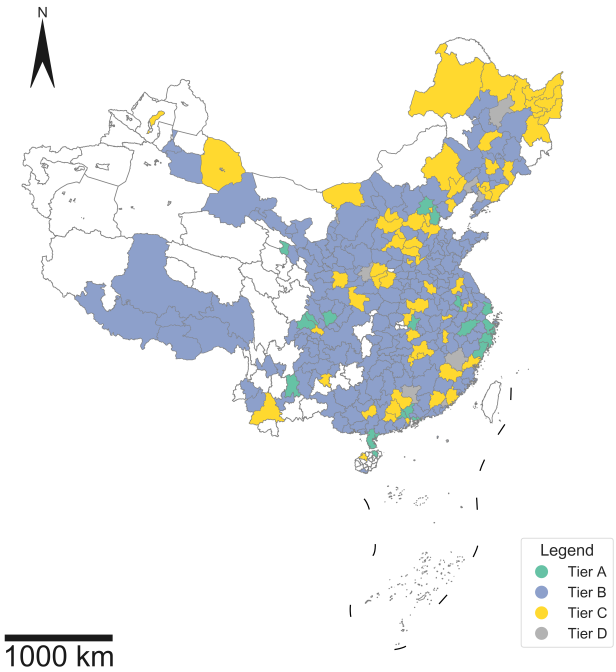


Figure 4. Geospatial distribution of the overall performance scores in the energy dimension. Cities in Tier A are marked in green, Tier B in blue, Tier C in yellow, and Tier D in grey. Blank regions indicate cities with no data.

Table 2. Distribution of cities across performance tiers by urban scale group.

City Group	Tier A	Tier B	Tier C	Tier D	Total
Mega (Group I)	11	10	0	0	21
Large (Group II)	7	62	10	1	80
Medium (Group III)	1	77	29	4	111
Small (Group IV)	0	59	22	3	84

Note: Sansha city was excluded from the performance tier analysis due to insufficient data availability.

A comparative analysis of performance across city groups was conducted using a pie chart distribution (Figure 5).

The results indicate that all Mega cities are positioned within Tiers A and B, while Large cities are primarily situated in Tier B. Medium and Small cities also predominantly occupy Tier B, though a notable proportion falls into Tier C. These findings suggest a positive correlation between urban economic scale and low-carbon infrastructure maturity, which is a key component of urban sustainability.

The proportion of city scales within each performance tier was further examined, as shown in Figure 6.

Tier A is composed almost exclusively of mega and large cities, with no representation from small cities. Conversely, Tiers C and D consist entirely of cities below the mega-city scale. This pattern suggests that while a high level of economic development provides a robust baseline for low-carbon energy construction, lower economic capacity may constrain the upper potential for such initiatives. Consequently, the progress of low-carbon development is inextricably linked to economic factors.

To elucidate the disparities in low-carbon energy construction across city scales, a raincloud plot was utilized [26] (Figure 7), combining statistical distributions with box plots to highlight group characteristics.

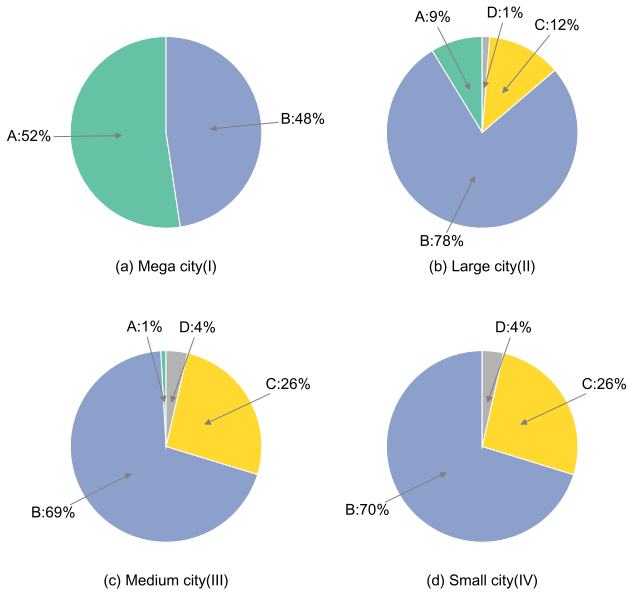


Figure 5. Pie chart of the distribution of cities in different performance tiers across all city scale groups: (**a**) Pie chart of the distribution of cities in Mega city group. Cities in Tier A are marked in green, Tier B in blue; (**b**) Pie chart of the distribution of cities in Large city group. Cities in Tier A are marked in green, Tier B in blue, Tier C in yellow, and Tier D in grey; (**c**) Pie chart of the distribution of cities in Medium city group. Cities in Tier A are marked in green, Tier B in blue, Tier C in yellow, and Tier D in grey; (**d**) Pie chart of the distribution of cities in Small city group. Cities in Tier B are marked in blue, Tier C in yellow, and Tier D in grey.

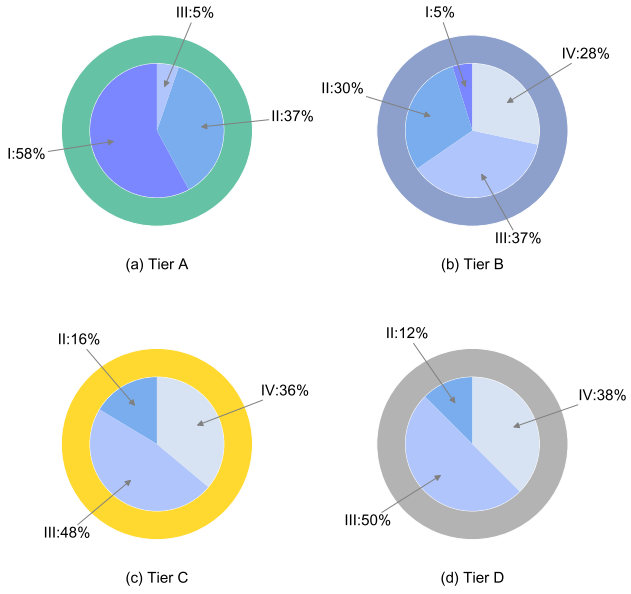


Figure 6. Pie chart of the distribution of cities in different performance tiers across all city tiers groups: (**a**) Pie chart of the distribution of cities in Tier A. Cities in Tier A are marked in green. And I stands for mega cities, II stands for large cities, III stands for medium cities. The depth of the blue color indicates the scale of the city; (**b**) Pie chart of the distribution of cities in Tier B. Cities in Tier B are marked in blue. And I stands for mega cities, II stands for large cities, III stands for medium cities, IV stands for small cities. The depth of the blue color indicates the scale of the city; (**c**) Pie chart of the distribution of cities in Tier C. Cities in Tier C are marked in yellow. And II stands for large cities, III stands for medium cities, IV stands for small cities. The depth of the blue color indicates the scale of the city; (**d**) Pie chart of the distribution of cities in Tier D. Cities in Tier D are marked in grey. And II stands for large cities, III stands for medium cities, IV stands for small cities. The depth of the blue color indicates the scale of the city.

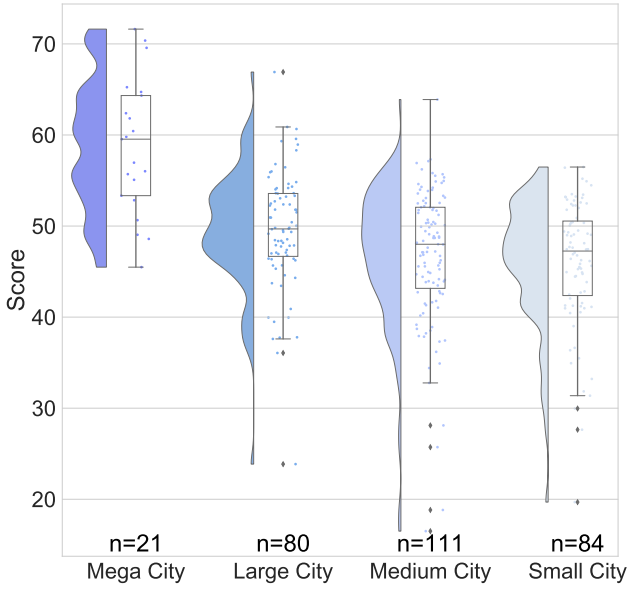


Figure 7. Raincloud plot of the distribution of cities in different scale groups. The depth of the blue color indicates the scale of the city.

The mean score for Mega-cities is significantly higher than that of other groups, whereas differences among the remaining groups are not statistically significant. This suggests a non-linear relationship between economic development and low-carbon construction, where substantial positive impacts manifest only after a certain economic threshold is reached. This non-linear dynamic is a critical consideration for targeted sustainability policy interventions.

Furthermore, considerable intra-group variation exists within the Mega-city category, indicating that economic prosperity does not automatically translate into superior low-carbon performance; urban governance and strategic positioning are equally vital. In contrast, the distributions for Medium and Small cities approximate a normal distribution but exhibit a pronounced long-tail effect, with several cities deviating significantly from their group averages.

4.2. Phase-Specific Performance Analysis

The characteristics of each PICOF phase were analyzed by examining the overall score distributions (Figure 8).

The "Plan (P)" and "Feedback (F)" phases exhibit distributions that approximate normality. In contrast, the "Implement (I)", "Check (C)", and "Outcome (O)" phases display varying degrees of skewness. The left skew in the "Implement" phase suggests that most cities demonstrate relatively high performance in execution. However, the right skew in the "Check" and "Outcome" phases indicates that the majority of cities face challenges in monitoring and achieving measurable results, highlighting a significant gap in the sustainable policy cycle.

The performance of cities across various scales was further analyzed using a parallel coordinate plot (Figure 9).

The results reveal a consistent tendency across nearly all scales to prioritize the "Implement" phase while systematically neglecting the "Check" phase. Notably, even some mega-cities recorded "Check" scores near zero. This pattern suggests a systemic deficiency in China's low-carbon energy governance, where resources and efforts are disproportionately allocated to planning and execution at the expense of rigorous monitoring and verification, posing a risk to the long-term sustainability of these efforts.

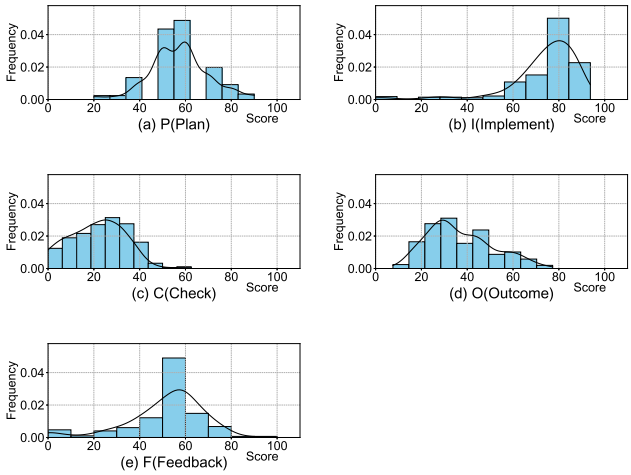


Figure 8. The distribution of all cities in different phases: (**a**) The distribution of all cities in Plan phase; (**b**) The distribution of all cities in Implement phase; (**c**) The distribution of all cities in Check phase; (**d**) The distribution of all cities in Outcome phase; (**e**) The distribution of all cities in Feedback phase.

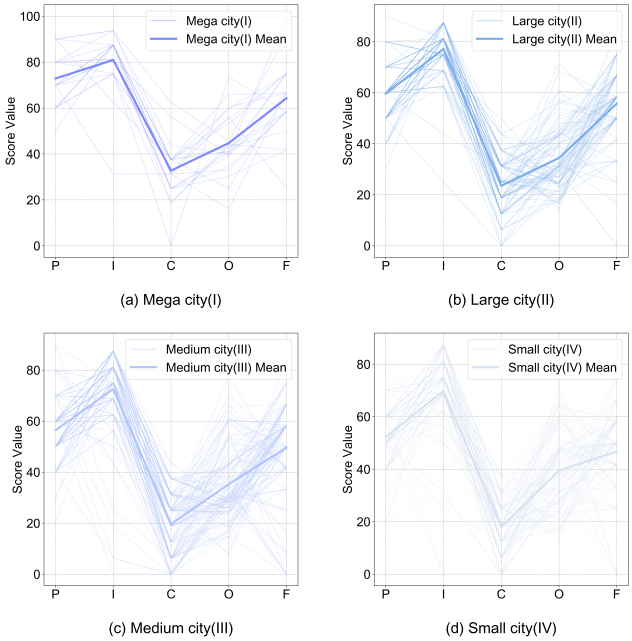


Figure 9. Parallel coordinate plot of the performance of cities in different scale groups: (**a**) Parallel coordinate plot of the performance of cities in Mega city group. The thinner lines represent the performance scores of individual cities, while the thicker lines indicate the average performance for each respective group; (**b**) Parallel coordinate plot of the performance of cities in Large city group. The thinner lines represent the performance scores of individual cities, while the thicker lines indicate the average performance for each respective group; (**c**) Parallel coordinate plot of the performance of cities in Medium city group. The thinner lines represent the performance scores of individual cities, while the thicker lines indicate the average performance for each respective group; (**d**) Parallel coordinate plot of the performance of cities in Small city group. The thinner lines represent the performance scores of individual cities, while the thicker lines indicate the average performance for each respective group.

A comparative analysis of performance across tiers within each phase was also conducted (Figure 10). While all tiers tend to overlook the "Check" phase, clear distinctions emerge in absolute scores. As cities descend from Tier A to Tier D, the relative emphasis on the "Feedback" phase decreases. In Tier D, cities often score lower in "Feedback" than in "Outcome," fundamentally altering their performance profiles. This indicates that the

relative prioritization of the "Feedback" phase serves as a proxy for the maturity of a city's low-carbon governance. The strong positive correlation observed between a city's performance tier and its "Feedback" score provides empirical evidence that robust feedback loops are essential for high-performing sustainable governance.

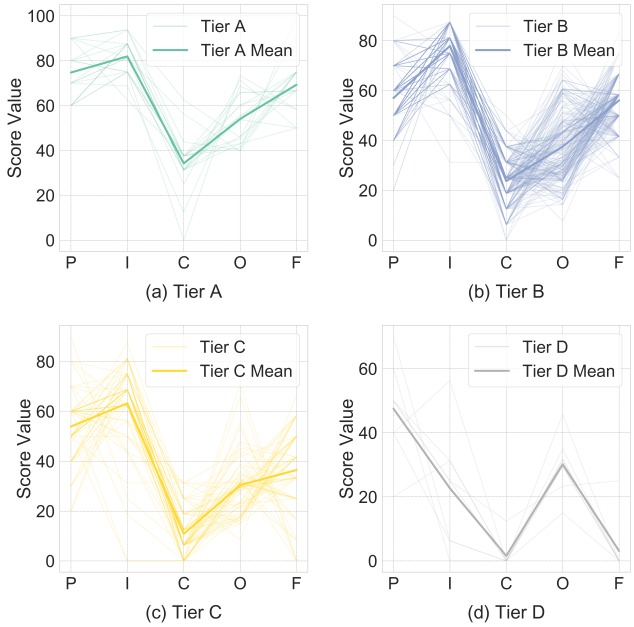


Figure 10. Parallel coordinate plot of the performance of cities in different tiers groups: (**a**) Parallel coordinate plot of the performance of cities in Tier A. The thinner lines represent the performance scores of individual cities, while the thicker lines indicate the average performance for each respective group; (**b**) Parallel coordinate plot of the performance of cities in Tier B. The thinner lines represent the performance scores of individual cities, while the thicker lines indicate the average performance for each respective group; (**c**) Parallel coordinate plot of the performance of cities in Tier C. The thinner lines represent the performance scores of individual cities, while the thicker lines indicate the average performance for each respective group; (**d**) Parallel coordinate plot of the performance of cities in Tier D. The thinner lines represent the performance scores of individual cities, while the thicker lines indicate the average performance for each respective group.

5. Discussion and Conclusions

This study leverages a novel RAG-based framework to conduct a comprehensive evaluation of the low-carbon energy construction in 296 Chinese cities. The findings not only reveal distinct performance patterns but also offer deep insights into the systemic strengths and weaknesses of current efforts.

5.1. Principal Findings and Interpretation

This study's evaluation of 296 Chinese cities reveals several critical findings relevant to urban sustainability. Our analysis reveals a strong, albeit non-linear, correlation between a city's scale and its low-carbon performance. The 19 Tier A cities are predominantly economically advanced coastal hubs, underscoring the critical role of economic strength in enabling ambitious low-carbon transitions. However, the significant performance variation observed within the Mega-city group, as shown in the raincloud plot (Figure 7), indicates that economic prowess is a necessary but not sufficient condition; effective urban governance and strategic commitment are crucial for translating economic advantages into tangible sustainable outcomes. For instance, comparison between high-performing and low-performing mega-cities reveals that despite similar economic foundations, high performers, such as Beijing, exhibit significantly more robust "Feedback" mechanisms

and stricter "Check" processes. In contrast, lower-performing peers, such as Chongqing, often lack these closed-loop governance structures, leading to policy fragmentation. In contrast, the performance of Medium and Small cities clusters more tightly around a lower mean, suggesting that developmental constraints may impose a ceiling on their low-carbon achievements.

The PICOF framework provides a granular view of the policy process, uncovering a critical, systemic bottleneck. The parallel coordinate plots (Figures 9 and 10) clearly show that cities across all scales and performance tiers tend to neglect the "Check" phase. This widespread oversight suggests that while cities are adept at planning and implementation, they lack robust mechanisms for monitoring and verifying the efficacy of their actions. This is a significant weakness in the policy cycle, as without proper checking, it is impossible to learn, adapt, and ensure that implemented measures are delivering the desired outcomes, which is essential for the long-term sustainability of climate initiatives.

Furthermore, the "Feedback" phase emerged as a key differentiator of governance maturity. In higher-tier cities, the "Feedback" score is relatively strong, indicating a more mature, iterative approach to governance. These cities demonstrate stronger feedback loops, enabling iterative policy refinement. Conversely, in lower-tier cities, particularly Tier D, the "Feedback" score is exceptionally low, sometimes even falling below the "Outcome" score. This suggests a more linear, "fire-and-forget" approach to policy, where the cycle often terminates after implementation, hindering long-term progress and learning in sustainable urban development.

Thanks to the RAG methodology, which allows us to analyze the textual evidence used for scoring, we can delve deeper into the substance of policy planning. The word cloud analysis of the "Plan" phase (Figure 11) reveals distinct strategic priorities linked to city scale. To provide quantitative evidence for these scale-related differences, the specific frequencies and proportions of key terms across the four city groups are detailed in Table 3. Apart from universal themes like "Energy" and "Green," mega and large cities prioritize strategic, forward-looking concepts like "development" and "low-carbon." This reflects their role as regional leaders tasked with balancing growth and sustainability while pioneering advanced technological solutions. Their discourse is one of leadership and transformation. In contrast, medium and small cities focus more on foundational concepts like "planning" and "ecology," reflecting administrative realities where priorities center on adhering to national directives and addressing localized environmental issues. This divergence highlights that cities are pursuing qualitatively different low-carbon development pathways shaped by their unique resource endowments and administrative capacities, all of which influence their long-term sustainability trajectories.

Table 3. Frequency of top 20 policy terms by city scale.

Rank	Mega City		Large City		Medium City		Small City	
	Term	**Freq.**	**Term**	**Freq.**	**Term**	**Freq.**	**Term**	**Freq.**
1	Energy	0.1783	Energy	0.1414	Energy	0.1027	Energy	0.0958
2	Green	0.0883	Green	0.0781	Green	0.0749	Green	0.0814
3	Low-carbon	0.0842	Low-carbon	0.0646	Planning	0.0687	Ecological	0.0755
4	Planning	0.0724	Development	0.0609	Ecological	0.0677	Planning	0.0658
5	Development	0.0669	Advance	0.0588	Low-carbon	0.0612	Development	0.0608
6	Construction	0.0518	Planning	0.0586	Advance	0.0583	Advance	0.0592
7	Advance	0.0496	Construction	0.0566	Development	0.0566	Construction	0.0557
8	Binding	0.0414	Ecological	0.0493	Construction	0.0550	Low-carbon	0.0545
9	Promote	0.0407	Emissions	0.0452	Emissions	0.0467	Emissions	0.0410
10	Emissions	0.0400	Clean	0.0424	Binding	0.0433	Clean	0.0389
11	Clean	0.0399	Key	0.0386	Implementation	0.0368	Binding	0.0382
12	Energy Conservation	0.0390	Binding	0.0378	Key	0.0359	Implementation	0.0378

Table 3. *Cont.*

Rank	Mega City		Large City		Medium City		Small City	
	Term	**Freq.**	**Term**	**Freq.**	**Term**	**Freq.**	**Term**	**Freq.**
13	Key	0.0370	Energy Conservation	0.0355	Target/Goal	0.0326	Target/Goal	0.0357
14	Target/Goal	0.0357	Implementation	0.0353	Promote	0.0324	Key	0.0352
15	System	0.0356	Promote	0.0351	Clean	0.0319	Strengthen	0.0341
16	Consumption	0.0345	Accelerate	0.0335	System	0.0317	Industry	0.0331
17	Implementation	0.0342	Target/Goal	0.0329	Strengthen	0.0311	Promote	0.0320
18	Ecological	0.0336	Strengthen	0.0329	14th Five-Year Plan	0.0302	Accelerate	0.0312
19	Strengthen	0.0324	Consumption	0.0311	Accelerate	0.0294	14th Five-Year Plan	0.0304
20	14th Five-Year Plan	0.0318	Industry	0.0299	Project	0.0288	System	0.0301

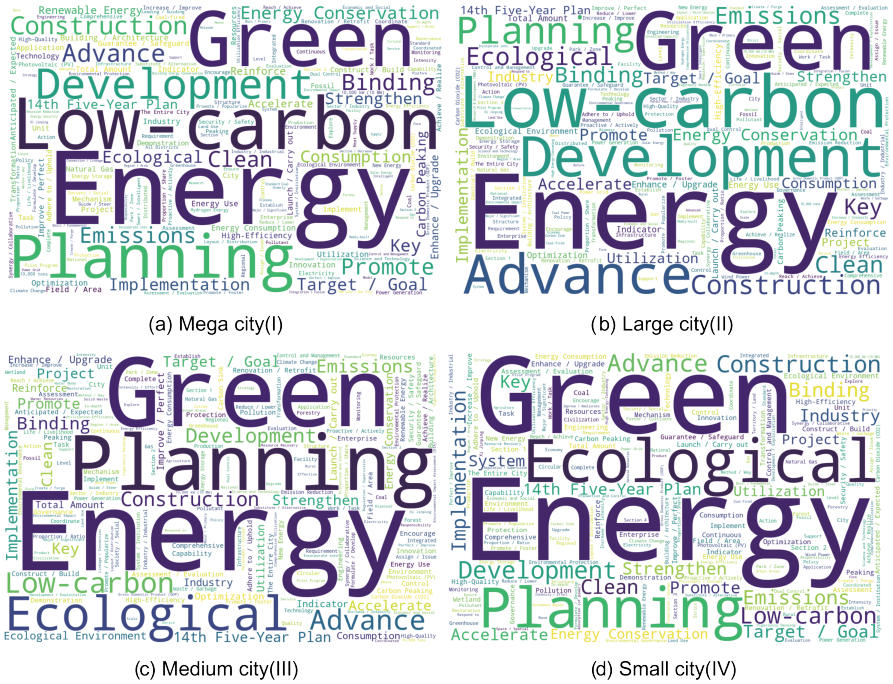


Figure 11. Wordcloud of the policy documents in the P (Plan) phase for cities in different scale groups: (**a**) Wordcloud of the policy documents in the P (Plan) phase for cities in Mega city group. The size of the word indicates its frequency within the retrieved policy texts; (**b**) Wordcloud of the policy documents in the P (Plan) phase for cities in Large city group. The size of the word indicates its frequency within the retrieved policy texts; (**c**) Wordcloud of the policy documents in the P (Plan) phase for cities in Medium city group. The size of the word indicates its frequency within the retrieved policy texts; (**d**) Wordcloud of the policy documents in the P (Plan) phase for cities in Small city group. The size of the word indicates its frequency within the retrieved policy texts.

5.2. Methodological Contributions and Policy Implications

The primary methodological contribution of this paper is the development and validation of a RAG-based PICOF framework for policy evaluation. Traditional methods often rely on static, outcome-based indicators, which can obscure the underlying reasons for success or failure. The PICOF cycle, by contrast, dissects the entire policy process, providing a dynamic and actionable diagnosis of where a city's efforts are succeeding or faltering. It shifts the focus from "what" was achieved to "how" and "why," enabling the identification of specific bottlenecks like the systemic weakness in the "Check" phase. This 'glass-box' approach offers a significant advancement over traditional methods by providing a transparent, scalable, and process-oriented assessment.

The integration of Retrieval-Augmented Generation (RAG) with a Large Language Model (LLM) enhances the objectivity, transparency, and depth of the evaluation. Unlike opaque expert scoring or superficial keyword analysis, our framework grounds its

assessment in concrete, verifiable evidence retrieved from source documents. This "glass-box" approach not only improves the credibility of the results but also enables deeper qualitative insights, as exemplified by our word cloud analysis, thereby providing a more robust foundation for evidence-based sustainability management. For example, when evaluating the 'Check' phase for a specific city, the system retrieved a text chunk stating "strictly implement the energy conservation target responsibility system and assessment mechanism." Based on this evidence, the LLM generated a high score for the supervision variable, transparently linking the policy language to the evaluation outcome.

The findings yield several policy implications. The universal neglect of the "Check" phase calls for national-level action to establish standardized Monitoring, Reporting, and Verification (MRV) systems. The divergence in performance underscores the need for differentiated, tailored policies rather than a one-size-fits-all approach. Finally, local policymakers, particularly in smaller cities, must prioritize building robust "Feedback" loops to foster continuous learning and adaptation, ensuring that urban governance remains responsive to evolving sustainability challenges.

5.3. Limitations

This study is subject to several limitations. First, its reliance on publicly available documents means that policy implementation realities may differ from official reports; information not in the public domain is not captured. This reliance may introduce a selection bias, where cities with more transparent or extensive documentation appear more effective than those with less digital presence. To mitigate this in future studies, the RAG-based analysis could be triangulated with other data sources, such as on-site surveys or third-party audit reports, to cross-verify the document-based findings. Second, while the RAG framework enhances objectivity, the LLM-based scoring is not entirely immune to inherent biases from its training data. Third, the analysis is confined to the energy sector, excluding other critical areas like transportation and industry. Lastly, the data collection process, though extensive, may have missed some relevant documents, potentially leading to an incomplete information base for certain cities, which may affect the granularity of the sustainability assessment.

5.4. Future Research

Future research should build on this work in several directions. The framework can be expanded to other critical sectors, such as transport and industry, to provide a more holistic view of urban sustainability transitions. While the core RAG architecture is domain-agnostic, scaling to these new sectors will require constructing domain-specific knowledge bases and fine-tuning the prompt engineering to accommodate distinct terminologies and policy frameworks (e.g., emission standards in transport vs. energy efficiency in industry). Incorporating non-public data through surveys or interviews could offer a more comprehensive understanding of on-the-ground implementation. Further refinement of the LLM evaluation protocol, including cross-model validation and advanced prompt engineering, could continue to enhance the framework's objectivity and accuracy. Finally, longitudinal analysis using this framework could track policy evolution over time, providing invaluable insights into the dynamics of urban governance and long-term sustainability.

Moreover, this framework holds significant potential for international replicability. While this study focuses on China, the RAG-based architecture is language-agnostic, provided that an appropriate embedding model and LLM are selected. Future applications in other countries should adapt the keyword search strategy to local policy terminologies and refine the PICOF prompts to align with specific national governance structures.

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Data Availability Statement: The data presented in this study are available on request from the corresponding author. The code and evaluation metrics developed for this study are openly available in a GitHub repository at https://github.com/jjwwczy/AutoRAG (accessed on 10 February 2026). Additionally, the core RAG prompt templates used for the PICOF phases are included in the repository to facilitate reproducibility. The full processed policy document corpus is not publicly shared due to copyright and privacy restrictions but can be obtained from the corresponding author upon reasonable request.

Conflicts of Interest: The authors declare no conflicts of interest.

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